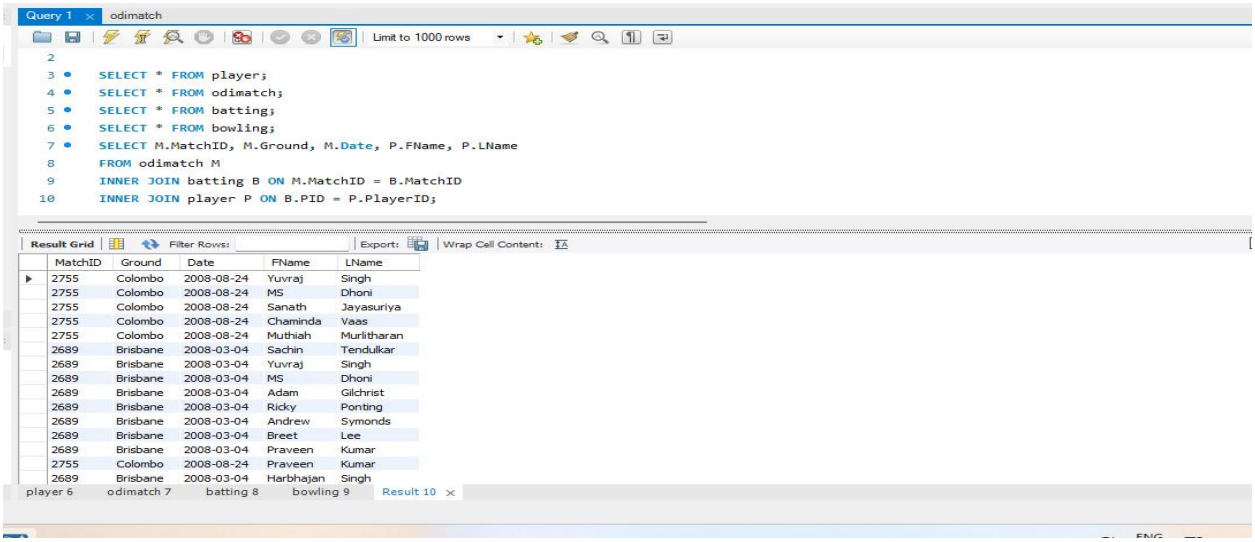
DBMS LAB 6

Experiment 5: Implementation of Various Join Operations

Aim: Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.

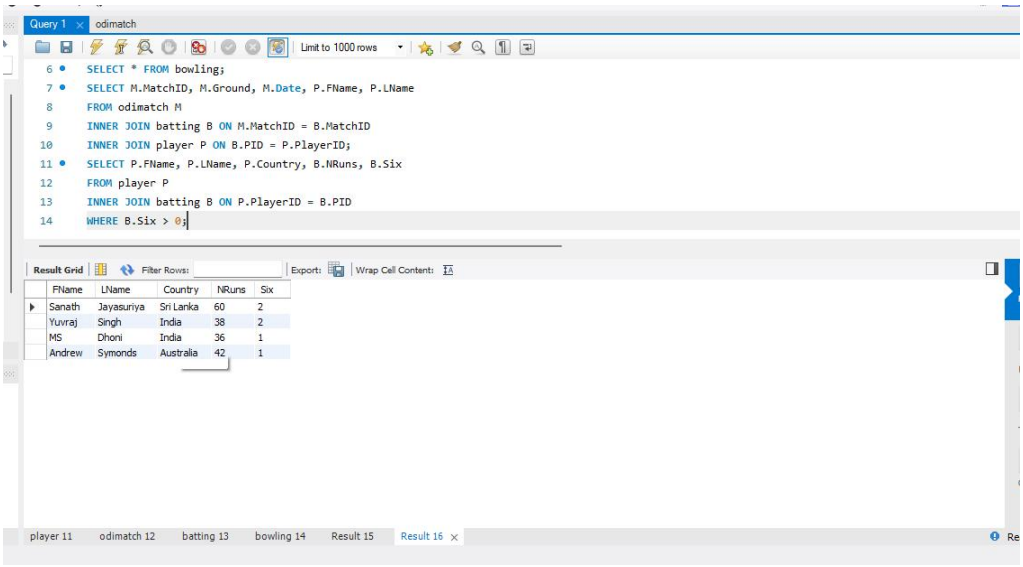
1 [Inner Join]: Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

```
SELECT M.MatchID, M.Ground, M.Date, P.FName, P.LName
FROM odimatch M
INNER JOIN batting B ON M.MatchID = B.MatchID
INNER JOIN player P ON B.PID = P.PlayerID;
```



2 [Inner Join with Filter]: Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

```
SELECT P.FName, P.LName, P.Country, B.NRuns, B.Six
FROM player P
INNER JOIN batting B ON P.PlayerID = B.PID
WHERE B.Six > 0;
```



3 [Left Outer Join]: Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

SELECT P.PlayerID, P.FName, P.LName, B.MatchID, B.NRuns

FROM player P

LEFT JOIN batting B ON P.PlayerID = B.PID;

Query 1 odimatch SQL File 4*

Limit to 1000 rows

```
1  USE mahek01;
2  SELECT P.PlayerID, P.FName, P.LName, B.MatchID, B.NRuns
3  FROM player P
4  LEFT JOIN batting B ON P.PlayerID = B.PID;
```

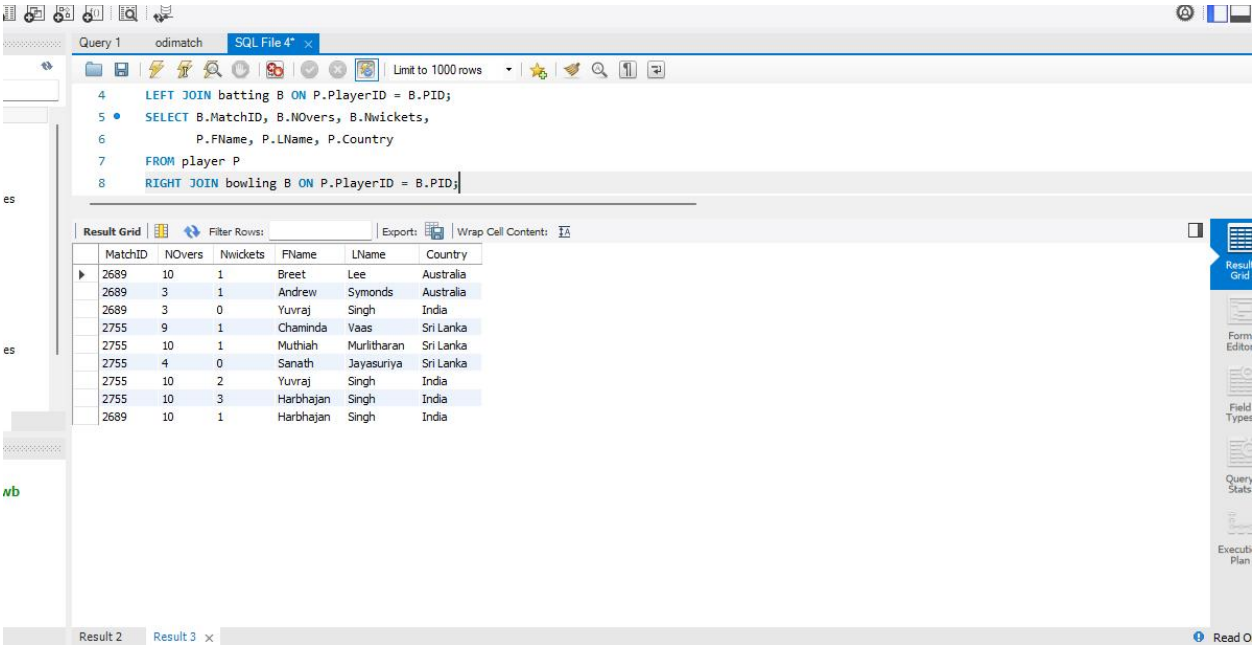
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PlayerID	FName	LName	MatchID	NRuns
21001	Virender	Sehwag	NULL	NULL
23001	Yuvraj	Singh	2755	0
23001	Yuvraj	Singh	2689	38
24001	Andrew	Symonds	2689	42
25001	MS	Dhoni	2755	71
25001	MS	Dhoni	2689	36
27001	Praveen	Kumar	2689	7
27001	Praveen	Kumar	2755	2
27002	Ishant	Sharma	NULL	NULL
89001	Sachin	Tendulkar	2689	91
90001	Brain	Lara	NULL	NULL
90002	Anil	Kumble	NULL	NULL
91001	Sanath	Jayasuriya	2755	60
92001	Shane	Warne	NULL	NULL
92002	Muthiah	Murlitharan	2755	1
92003	Inzamam	UI-Huq	NULL	NULL
93002	Heath	Streak	NULL	NULL
93003	Gary	Kirsten	NULL	NULL
94002	Chaminda	Vaas	2755	17
94004	Stephen	Fleming	NULL	NULL

Result 1

4 [Right Outer Join]: Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

```
SELECT B.MatchID, B.NOvers, B.Nwickets,
    P.FName, P.LName, P.Country
FROM player P
RIGHT JOIN bowling B ON P.PlayerID = B.PID;
```



5 [Full Outer Join / Union]: Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

```
SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
FROM batting B
LEFT JOIN bowling W
ON B.MatchID = W.MatchID AND B.PID = W.PID
UNION
SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
FROM batting B
RIGHT JOIN bowling W
ON B.MatchID = W.MatchID AND B.PID = W.PID;
```

Query 1 odimatch SQL File 4* Limit to 1000 rows

```
9
10 •  SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
11    FROM batting B
12    LEFT JOIN bowling W
13    ON B.MatchID = W.MatchID AND B.PID = W.PID
14    UNION
15    SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
16    FROM batting B
17    RIGHT JOIN bowling W
18    ON B.MatchID = W.MatchID AND B.PID = W.PID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MatchID	PID	NRuns	NOvers	Nwickets
2755	23001	0	10	2
2755	25001	71	NULL	NULL
2755	91001	60	4	0
2755	94002	17	9	1
2755	92002	1	10	1
2689	89001	91	NULL	NULL
2689	23001	38	3	0
2689	25001	36	NULL	NULL
2689	99002	2	NULL	NULL
2689	95001	1	NULL	NULL
2689	24001	42	3	1
2689	99001	7	10	1
2689	27001	7	NULL	NULL

Result 4 Result 5 Result 6

6 [3-Table Inner Join]: List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

```
SELECT P.FName, P.LName, M.Ground, M.Date, B.NRuns
FROM player P
INNER JOIN batting B ON P.PlayerID = B.PID
INNER JOIN odimatch M ON B.MatchID = M.MatchID
WHERE M.Team1 = 'India' OR M.Team2 = 'India';
```

Query 1 odimatch SQL File 4*

Limit to 1000 rows

```
1  USE mahek01;
2  SELECT P.FName, P.LName, M.Ground, M.Date, B.NRuns
3  FROM player P
4  INNER JOIN batting B ON P.PlayerID = B.PID
5  INNER JOIN odimatch M ON B.MatchID = M.MatchID
6  WHERE M.Team1 = 'India' OR M.Team2 = 'India';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FName	LName	Ground	Date	NRuns
Yuvraj	Singh	Colombo	2008-08-24	0
MS	Dhoni	Colombo	2008-08-24	71
Sanath	Jayasuriya	Colombo	2008-08-24	60
Chaminda	Vaas	Colombo	2008-08-24	17
Muthiah	Murlitharan	Colombo	2008-08-24	1
Sachin	Tendulkar	Brisbane	2008-03-04	91
Yuvraj	Singh	Brisbane	2008-03-04	38
MS	Dhoni	Brisbane	2008-03-04	36
Adam	Gilchrist	Brisbane	2008-03-04	2
Ricky	Ponting	Brisbane	2008-03-04	1
Andrew	Symonds	Brisbane	2008-03-04	42
Breet	Lee	Brisbane	2008-03-04	7
Praveen	Kumar	Brisbane	2008-03-04	7
Praveen	Kumar	Colombo	2008-08-24	2
Harbhajan	Singh	Brisbane	2008-03-04	3
Harbhajan	Singh	Colombo	2008-08-24	2

Result 7 Result 8 Result 9 Result 10

7 [4-Table Multi-Join]: Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

```
SELECT P.FName, P.LName, M.Ground, M.Date,
    B.NRuns, W.Nwickets
FROM player P
INNER JOIN batting B ON P.PlayerID = B.PID
INNER JOIN bowling W
ON B.MatchID = W.MatchID AND B.PID = W.PID
INNER JOIN odimatch M ON B.MatchID = M.MatchID;
```

Query 1 odimatch SQL File 4*

Limit to 1000 rows

```
1  USE mahek01;
2  SELECT P.FName, P.LName, M.Ground, M.Date,
3         B.NRuns, W.Nwickets
4  FROM player P
5  INNER JOIN batting B ON P.PlayerID = B.PID
6  INNER JOIN bowling W
7  ON B.MatchID = W.MatchID AND B.PID = W.PID
8  INNER JOIN odimatch M ON B.MatchID = M.MatchID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FName	LName	Ground	Date	NRuns	Nwickets
Breet	Lee	Brisbane	2008-03-04	7	1
Andrew	Symonds	Brisbane	2008-03-04	42	1
Yuvraj	Singh	Brisbane	2008-03-04	38	0
Chaminda	Vaas	Colombo	2008-08-24	17	1
Muthiah	Murlitharan	Colombo	2008-08-24	1	1
Sanath	Jayasuriya	Colombo	2008-08-24	60	0
Yuvraj	Singh	Colombo	2008-08-24	0	2
Harbhajan	Singh	Colombo	2008-08-24	2	3
Harbhajan	Singh	Brisbane	2008-03-04	3	1

Result 11

8 [Self Join - Attribute Match]: Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

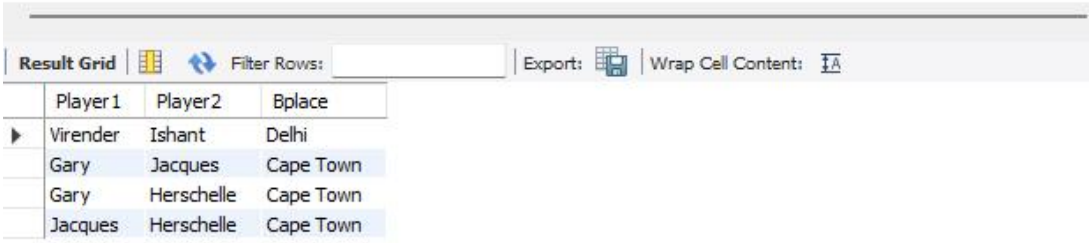
SELECT P1.FName AS Player1, P2.FName AS Player2, P1.Bplace

FROM player P1

INNER JOIN player P2 ON P1.Bplace = P2.Bplace

WHERE P1.PlayerID < P2.PlayerID;





Player1	Player2	Bplace
Virender	Ishant	Delhi
Gary	Jacques	Cape Town
Gary	Herschelle	Cape Town
Jacques	Herschelle	Cape Town

9 [Self Join - Match Analysis]: Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

```
SELECT M1.MatchID AS Match1, M2.MatchID AS Match2,
    M1.Ground, M1.Date AS Date1, M2.Date AS Date2
FROM odimatch M1
INNER JOIN odimatch M2 ON M1.Ground = M2.Ground
WHERE M1.Date <> M2.Date
AND M1.MatchID < M2.MatchID;
```

Query 1 odimatch SQL File 4*

Limit to 1000 rows

```
1  USE mahek01;
2  SELECT M1.MatchID AS Match1, M2.MatchID AS Match2,
3         M1.Ground, M1.Date AS Date1, M2.Date AS Date2
4  FROM odimatch M1
5  INNER JOIN odimatch M2 ON M1.Ground = M2.Ground
6  WHERE M1.Date <> M2.Date
7  AND M1.MatchID < M2.MatchID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Match1	Match2	Ground	Date1	Date2
2404	2750	Colombo	2006-08-16	2008-08-24
2404	2755	Colombo	2006-08-16	2008-08-24

Result 13

10 [Cross Join]: Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

```
SELECT P.FName, P.LName, M.MatchID, M.Ground, M.Date
FROM player P
CROSS JOIN odimatch M
WHERE P.Country = 'India';
```

Query 1 odimatch SQL File 4*

Limit to 1000 rows

```
1  USE mahek01;
2  SELECT P.FName, P.LName, M.MatchID, M.Ground, M.Date
3  FROM player P
4  CROSS JOIN odimatch M
5  WHERE P.Country = 'India';
```

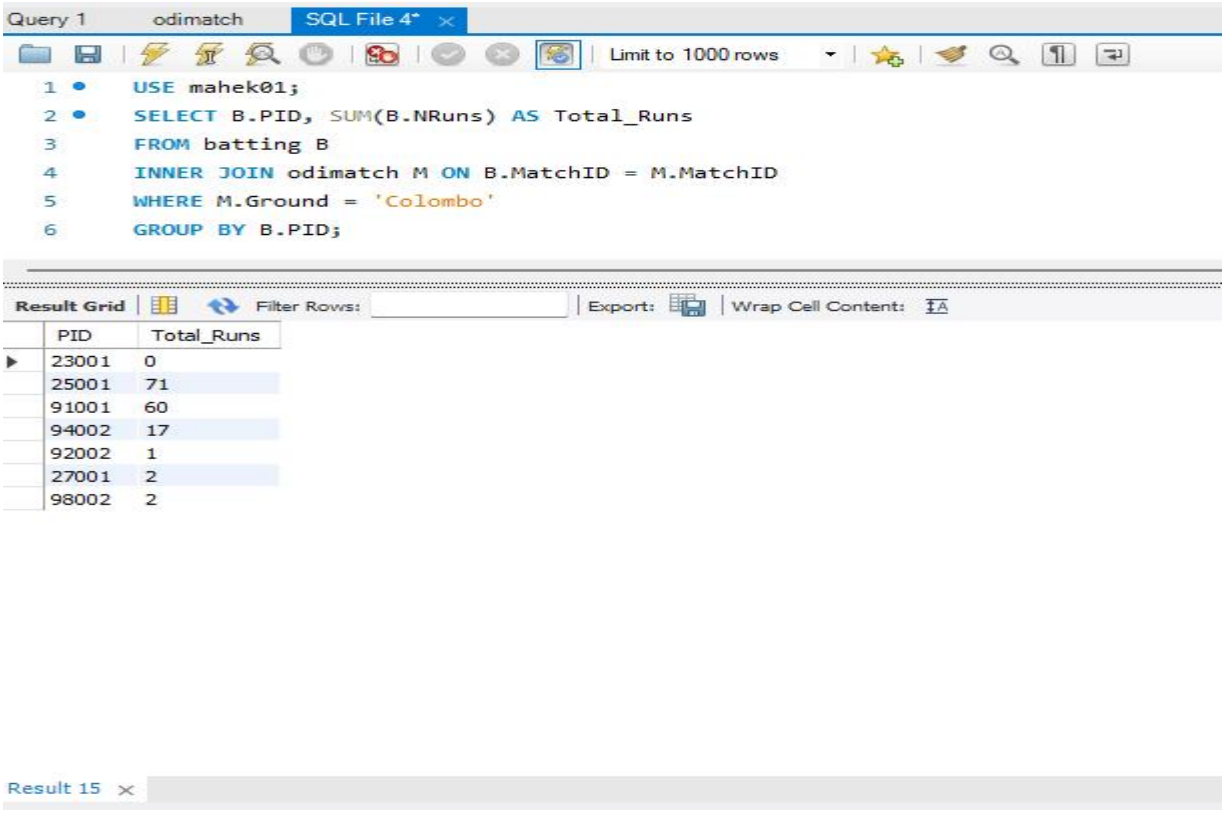
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FName	LName	MatchID	Ground	Date
Sam	NULL	2324	Peshwar	2006-02-06
Harbhajan	Singh	2324	Peshwar	2006-02-06
Saurav	Ganguly	2324	Peshwar	2006-02-06
Rahul	Dravid	2324	Peshwar	2006-02-06
Anil	Kumble	2324	Peshwar	2006-02-06
Sachin	Tendulkar	2324	Peshwar	2006-02-06
Ishant	Sharma	2324	Peshwar	2006-02-06
Praveen	Kumar	2324	Peshwar	2006-02-06
MS	Dhoni	2324	Peshwar	2006-02-06
Yuvraj	Singh	2324	Peshwar	2006-02-06
Virender	Sehwag	2324	Peshwar	2006-02-06
Sam	NULL	2327	Rawalpi...	2006-02-11
Harbhajan	Singh	2327	Rawalpi...	2006-02-11
Saurav	Ganguly	2327	Rawalpi...	2006-02-11
Rahul	Dravid	2327	Rawalpi...	2006-02-11
Anil	Kumble	2327	Rawalpi...	2006-02-11
Sachin	Tendulkar	2327	Rawalpi...	2006-02-11
Ishant	Sharma	2327	Rawalpi...	2006-02-11
Praveen	Kumar	2327	Rawalpi...	2006-02-11
MS	Dhoni	2327	Rawalpi...	2006-02-11

Result 14

11 [Natural Join vs Explicit]: Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

```
SELECT B.PID, SUM(B.NRuns) AS Total_Runs
FROM batting B
INNER JOIN odimatch M ON B.MatchID = M.MatchID
WHERE M.Ground = 'Colombo'
GROUP BY B.PID;
```



12 [Join with Aggregation]: Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

```
SELECT P.PlayerID, P.FName, P.LName,
    SUM(B.NRuns) AS Total_Runs
FROM player P
INNER JOIN batting B ON P.PlayerID = B.PID
GROUP BY P.PlayerID, P.FName, P.LName
ORDER BY Total_Runs DESC;
```

```
Query 1   odimatch   SQL File 4*
Limit to 1000 rows
1  USE mahek01;
2  SELECT P.PlayerID, P.FName, P.LName,
3         SUM(B.NRuns) AS Total_Runs
4  FROM player P
5  INNER JOIN batting B ON P.PlayerID = B.PID
6  GROUP BY P.PlayerID, P.FName, P.LName
7  ORDER BY Total_Runs DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PlayerID	FName	LName	Total_Runs
25001	MS	Dhoni	107
89001	Sachin	Tendulkar	91
91001	Sanath	Jayasuriya	60
24001	Andrew	Symonds	42
23001	Yuvraj	Singh	38
94002	Chaminda	Vaas	17
27001	Praveen	Kumar	9
99001	Breet	Lee	7
98002	Harbhajan	Singh	5
99002	Adam	Gilchrist	2
92002	Muthiah	Murlitharan	1
95001	Ricky	Ponting	1

Result 16

13 [Left Join Anti-Join]: Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

SELECT P.PlayerID, P.FName, P.LName, P.Country

FROM player P

LEFT JOIN bowling B ON P.PlayerID = B.PID

WHERE B.PID IS NULL;

```
Query 1   odimatch   SQL File 4*
Limit to 1000 rows
1  USE mahek01;
2  SELECT P.PlayerID, P.FName, P.LName, P.Country
3  FROM player P
4  LEFT JOIN bowling B ON P.PlayerID = B.PID
5  WHERE B.PID IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan

Result 17